XF

PORTFOLIO · 2024–2026

Xinyue Feng

ALSO KNOWN AS LILY

Games, mods & interactive web

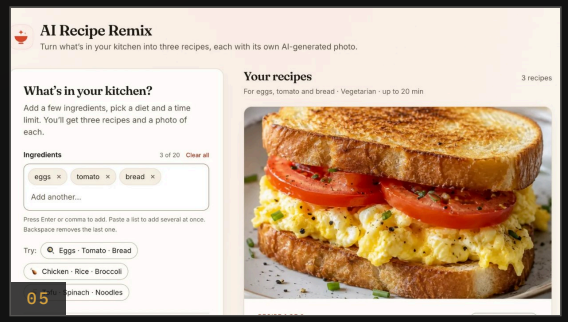
Computer Science student at Kean University, class of 2027. I make games and interactive web apps, and I care most about how they feel to use.

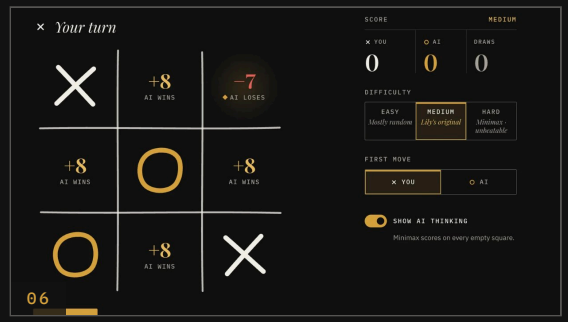












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About

I'm a Computer Science student at Kean University, graduating in 2027. I'm drawn to human–computer interaction and user-centered design, and I learn best by building things people can use and play.

The Java platformer I designed, drew and coded on my own grew out of a game I loved, *Ace Attorney*. Class assignments became projects too: a recipe app that turns what's in your fridge into three recipes with photos, and a tic-tac-toe game with a rule-based AI opponent. In summer 2026 I built a Harry Potter mod for Terraria and took it through route-by-route testing to a 1.0 release candidate.

In October 2026 I led two bigger projects with AI tools: Yuzu Onsen Capybaras, a cozy pixel-art life sim whose winter demo plays in the browser, and Otherworlds, a crossover boss mod for Terraria that is still in development.

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WEBSITE
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EDUCATION

Kean University — EXPECTED 2027
B.S. in Computer Science
Relevant coursework: HCI / UX, Software Engineering, Data Structures, Databases, Web Development.

EXPERIENCE

Chengdu Sheer Tianyi Game Technology — JUL 2026
TA Intern Engineer, Information Technology Department
Worked with Cascadeur 3D animation and its AI frame interpolation, and on a price-comparison crawler for third-party ticketing platforms.

Chengdu Sanying Technology — MAY – JUN 2026
Computer Vision Data Processing Intern
Organized image data and helped classify images as part of a computer-vision team.

Chengdu 2025 World Games Executive Committee — 2025
Sports Intern · Outstanding Intern
Coordinated competition logistics and communication between teams.

AWARDS & ACTIVITIES

Dean's Honor List, Kean University — SPRING 2026

The Honor Society of Phi Kappa Phi — MEMBER

CiGA Game Jam — 2024
Team member in a 48-hour jam: gameplay design and interaction flow for a playable prototype, refined through rounds of playtesting.

Lanqiao Cup, Third Prize — 2024
National Software & IT Competition, Visual Arts Design track, Zhejiang division.

TOOLBOX

Languages
Java · JavaScript / TypeScript · HTML & CSS

Frameworks
React · Tailwind CSS · Node.js & Express · Java Swing

Tools
Git & GitHub · REST APIs, OpenAI API · Aseprite · AI coding agents (Claude Code, Codex) · AI image generation

BEYOND CODE

I've volunteered as a tutor for children in rural areas and helped organize inclusive sports activities for autistic young people.

Selected work

2024–2026



01 · PIXEL-ART LIFE SIM · 2026

Yuzu Onsen Capybaras

A cozy life sim in a hot-spring town, led with AI tools. The winter demo plays in the browser.

04



02 · TERRARIA MOD · IN DEVELOPMENT

Otherworlds

38 boss fights from five games, rebuilt move by move for Terraria, led with AI tools.

08

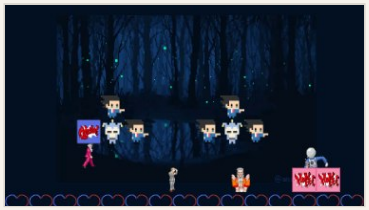


03 · TERRARIA MOD · 2026

Wizarding World

A Harry Potter adventure for Terraria: 12 boss fights and a Horcrux hunt.

11

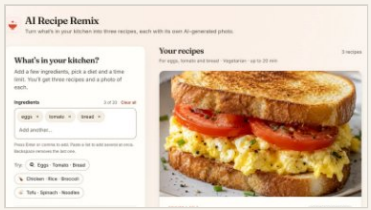


04 · JAVA GAME · 2024

The Adventures of Edgeworth

An Ace Attorney platformer written from scratch in Java Swing, now playable in the browser.

14



05 · FULL-STACK WEB · 2026

AI Recipe Remix

Three recipes and a photo of each dish from what's in your kitchen.

16



06 · GAME AI · 2026

Tic-Tac-Toe vs. AI

A rule-based AI and an unbeatable minimax search, side by side.

18

How the work was made: the Java game (2024), the recipe app and the tic-tac-toe AI (2026) are my coursework. I developed Wizarding World with AI tools and led the development of Yuzu Onsen Capybaras and Otherworlds with them, and the 2026 portfolio editions of the older projects were built with AI coding tools. Each project's pages say how.

01

PIXEL-ART LIFE SIM · 2026

Yuzu Onsen Capybaras

A cozy life sim in a hot-spring town

You help out at a capybara pet shop on the edge of a little hot-spring town in the mountains: look after seven capybaras, grow a garden, make friends around town and join a festival every week, through a first winter and then round the year.

ROLE
Lead developer with AI tools: direction, feedback and playtesting

STATUS
Free winter demo online; desktop app prepared for Steam, October 2026

PLATFORM
Web browsers on computers and phones; a Windows and macOS desktop app

BUILT WITH
JavaScript, HTML canvas, Web Audio, Electron, Python tooling; Claude Code and AI image generation

▶ Play the winter demo
capybara.sam-si.cn

Case study and trailer
samsi0322.github.io/xinyue-feng/work/yuzu-onsen-capybaras



The hot spring in the shop's yard, in the first winter.



What it is

The shop

The game is built around a capybara pet shop. Its seven capybaras each have a personality of their own and get on with their day whether you help or not: they nap, eat, swim among the pool floats, soak in the hot spring and sunbathe in the yard, with a small icon over each one's head to show how it feels. You feed them, pat them, name them and dress them up, while customers and regulars drop in, pay and leave their stars.

The town

Out of the shop's door is the town. Thirteen townsfolk keep days of their own, each with tastes, a birthday, letters and three stories told as a friendship grows, and six neighbors on Pine Lantern Street have stories of their own. There is a garden to grow, a forest to forage and fish in, a notice board of requests, an old bathhouse to bring back to life, and a book of wishes that changes the town as you fill it.

7 CAPYBARAS	**19** PEOPLE TO BEFRIEND	**16** FESTIVALS
18 CROPS	**30** CAVE FLOORS	**2** LANGUAGES



The shop at breakfast, with one capybara's menu open.

A year in town

Time runs on a calendar of four seasons, each with its own art, weather, crops, fish, recipes and sounds, and the townsfolk dress for each one. Every week brings a festival, sixteen in all, six with a game or contest of their own:

WINTER
Snow Lantern Night, a capybara fashion show, a yuzu bath and New Year’s Eve.

SPRING
A doll festival on the stream, a blossom picnic, a spring fair with goldfish scooping, and a capybara race on Children’s Day.

SUMMER
Tanabata wishes, blindfolded watermelon splitting, fireworks over the pond and paper lanterns on the stream.

AUTUMN
Moon viewing, catching falling leaves, a harvest festival with a prize for the finest vegetable, and a sweet-potato bonfire.



The garden in summer, grown from one bed to three.



Spring: the blossom picnic.



Summer: fireworks over the pond.



Autumn: the leaf viewing in the park.



Winter: Snow Lantern Night in the square.

Under the mountain, and ready to ship



A floor of crystals in the Spring Vein.



The source, thirty floors down.



Goldfish scooping at the spring fair.



▶ Watch the 92-second trailer, made from the game itself.

Under the mountain

Under the mountain shrine, the old spring keepers' tunnels go down thirty floors to the source of the town's hot water. You knock open rocks for ore, crystals and fossils, find each ladder by its wisp of steam, ride the steam lifts and fish the underground pools. There are no monsters: just the warm dark and your lantern.

The desk pet

A spin-off app puts one of the shop's capybaras on your desktop, a little like QQ Pet: it ambles along the bottom of the screen, and a toolbar gives it snacks, a hot bath, a paddling pool, a nap or the wardrobe. It runs on macOS and Windows.

Getting it ready to ship

- Keyboard and mouse, touch or a controller; English and Simplified Chinese; three save slots.
- A free web demo of the first winter. The full game unlocks with a key checked by a small Cloudflare Worker, and the demo build leaves out the other seasons' scenery, festivals, letters and dialogue.
- An Electron desktop app for Windows and macOS. The Windows build is prepared for Steam with 20 achievements, store art in both languages and the trailer; the Mac build waits on Apple's notarization. The Steam release is still to come.
- 74 automated test suites with more than 3,000 checks: scripted play in Chromium, WebKit and Firefox, screenshot comparisons, and mutation testing in which 720 planted bugs were all caught.

How it was made

Lily led the game's development with AI tools, credited openly. She built it with Claude Code, which wrote the JavaScript to her direction. The pixel art was generated with AI image tools, then cut out, shrunk onto a pixel grid and limited to a few colors each by the project's own scripts, and the text was written with the help of an AI language model. The music and sound effects are composed and synthesized by the game's own code as it plays.

Her feedback on the first version shaped its look and feel: a pixel font and pixel-style dialogue boxes, English by default, sound that works on iPhones, a little bed and a scarf for the capybaras.

02

TERRARIA MOD · IN DEVELOPMENT

Otherworlds

A crossover boss mod for Terraria

Bosses and gear from five other games come into Terraria: Hollow Knight, Hollow Knight: Silksong, Dead Cells, Cuphead and The Binding of Isaac. Each game is a pack that stands on its own and slots into Terraria's own progression.

ROLE
Lead developer with AI tools

STATUS
In development since October 2026; not yet released

PLATFORM
Terraria via tModLoader 1.4.4, .NET 8; single player and multiplayer

BUILT WITH
C#, Python tooling; Claude Code, OpenAI Codex and AI image generation

Case study
samsi0322.github.io/xinyue-feng/work/otherworlds



In-game screenshots: the Absolute Radiance, the Devil, Conjunctivius and Monstro.



Five games, thirty-eight bosses

Each pack is complete on its own terms and wired into Terraria's classes, progression and crafting: the weapons belong to Terraria's melee, ranged, magic and summoner classes, grow stronger as the world does, and are made from Terraria's ores and boss drops as well as the pack's own materials.

Every boss gets the full Terraria treatment: a summoning item, a boss bar, changes in Expert and Master Mode, a treasure bag, and a trophy, mask and relic.

38	5	2
BOSS FIGHTS	GAME PACKS	LANGUAGES

Hollow Knight · 13
From the False Knight to the Absolute Radiance

Silksong · 7
From the Bell Beast to the Bell Eater

Dead Cells · 7
From Mama Tick to the Queen

Cuphead · 9
From the Root Pack to the Devil, with the parry

The Binding of Isaac · 2
Monstro and the Duke of Flies, in a top-down basement of its own below a trapdoor

Fights true to the originals

Each boss starts as a written move list of the original: the wind-up, strike and recovery of every attack, how it picks its next move, and how a player dodges and hits back. The moves then come into Terraria one at a time. No boss is a single still image: every move has frames of its own.

In the game, and under test



The Absolute Radiance (Hollow Knight).



The Pure Vessel focuses.



Grand Mother Silk (Silksong).



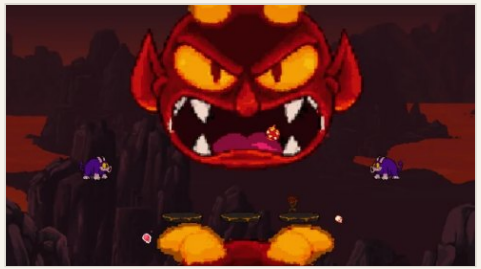
Conjunctivius (Dead Cells).



The Time Keeper lunges.



Sally Stageplay (Cuphead).



The Devil's giant head.



Monstro in the basement (Isaac).

Testing at scale

On a headless tModLoader server, a stand-in player fights every boss in a cleared sky arena, in Normal, Expert and Master Mode, with fixed random seeds, so any fight can be replayed exactly. The simulations check that a boss keeps after the player, never freezes, uses every move, dies on time and drops its loot. The same fights then run in generated worlds: 207 cases per seed for 33 bosses so far.

Seen as a player sees it

Scripted scenes in the game client photograph each fight, hundreds of frames at a time, and a motion recorder flags animation that jumps or stutters. 4,337 self-checks run inside the mod, and a change reaches the main line only after the build, the self-checks and the simulations pass.

How it is made

Lily is leading the mod with AI tools, credited openly: the C# is written by Claude Code and OpenAI's Codex, working in parallel branches under her direction, and the sprites are generated with AI image tools and fitted to Terraria's style by the project's scripts. The mod contains nothing from the original games: the bosses were studied from them, but every sprite is newly drawn and never traced.

An unofficial, non-commercial fan project, not affiliated with or endorsed by Team Cherry, Motion Twin, Evil Empire, StudioMDHR, Edmund McMillen, Nicalis or Re-Logic.

03

TERRARIA MOD · 2026

Wizarding World

A Harry Potter mod for Terraria

You start as a Hogwarts student with a first wand and follow a guided route through 12 boss fights and a Horcrux hunt to Lord Voldemort, the true final boss, then claim the Deathly Hallows in a short epilogue.

ROLE

Solo developer: design, code, art direction, testing and release

STATUS

1.0 release candidate, July 2026

PLATFORM

Terraria via tModLoader 1.4.4, .NET 8

BUILT WITH

C#, Python tooling, GitHub Actions; Claude Code and AI image generation

Source and player guide

github.com/SamSi0322/WizardingWorld

Case study

samsi0322.github.io/xinyue-feng/work/wizarding-world



Release artwork.



What it is

Wizarding World adds a complete Harry Potter progression to Terraria. The framing device is a Pensieve: you relive memories from across wizarding history, which explains why a first-year can meet the Basilisk, Aragog and the Battle of Hogwarts in one playthrough, and lets you replay any boss.

Around the bosses sit wand lore with 18 woods, 5 cores and 9 spell schools; a Dark Arts corruption meter; class quests; Quidditch seasons with Seeker ranks; Hogsmeade; and a three-zone Forbidden Forest.

12 BOSS FIGHTS	**24** ACTIVE WANDS	**78** ENEMIES
16 TOWN NPCS	**64** ACCESSORIES	**3** LANGUAGES



Platform Nine and Three-Quarters.

The boss route

- Mountain Troll
- Basilisk
- Fluffy
- Umbridge
- Bellatrix
- Dementor King
- Quirrell
- Aragog
- Horntail
- Fenrir
- Barty Crouch Jr.
- Lord Voldemort

The world, and getting it release-ready



Hogwarts across the Black Lake.



The Quidditch pitch.



The Forbidden Forest.

A mod this size (597 C# files, 624 sprites) breaks in quiet ways: a missing sprite, an untranslated line, a boss flag that doesn't survive a save. Most of the late work went into making that visible and testable:

- The main route was smoke-tested in tModLoader in ten stages, from the Hogwarts letter to the Hallows ending, each written up as a QA record.
- Key boss and story flags were confirmed to persist after Save & Exit.
- Python scripts count the content from the source and check the player guide against it: 665 checks in GitHub Actions.
- Debug commands are compiled out of public builds, and the release build has zero warnings.

Lily developed the mod on her own, with AI tools the repository credits openly: she wrote the C# with Claude Code and generated the sprites from written prompts kept in the repo. Harry Potter © Warner Bros. Entertainment and J.K. Rowling; Terraria © Re-Logic. A non-commercial fan project.



Poster art for the release.

04

JAVA GAME · CPS 2231 · SPRING 2024

The Adventures of Edgeworth

An Ace Attorney platformer

A Mario-style platformer written from scratch in Java Swing as a course project: six hand-designed levels, three kinds of enemy, a hidden block and an ending. It started as a desktop app; a 2026 port runs it in the browser.

ROLE
Solo: game design, level design, sprites, code (2024). The 2026 port was built with AI coding tools.

TIMELINE
Apr 26 – May 12, 2024

CONTEXT
CPS 2231 course project, Wenzhou-Kean University

BUILT WITH
Java 8, Swing, Java Sound · Eclipse, Aseprite, Excel

▶ Play in the browser
samsi0322.github.io/xinyue-feng/play/edgeworth

Case study
samsi0322.github.io/xinyue-feng/work/edgeworth



The web edition, 2026.

Levels, collisions and a web edition

The idea

Lily built the game around two things she loved: the little monster-adventure games on the 4399 web-game site she played as a child, and Capcom's *Ace Attorney*. So the hero is the prosecutor Miles Edgeworth, the enemies are courtroom regulars, and the blocks are the series' "Hold it!" and "Objection!" speech bubbles.



Miles Edgeworth walks, jumps, stomps.



Winston Payne paces; land on his head.



Manfred von Karma can't be stomped.



Wendy Oldbag rises out of pipes.



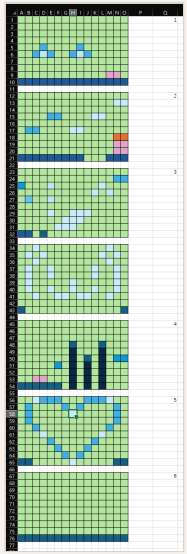
Dick Gumshoe points at a hidden block.

How it works

Nine classes, about 1,400 lines, no game engine: a double-buffered 900 × 600 window, and the player, every enemy and the end flag each on its own thread. Collision works on the tile grid:

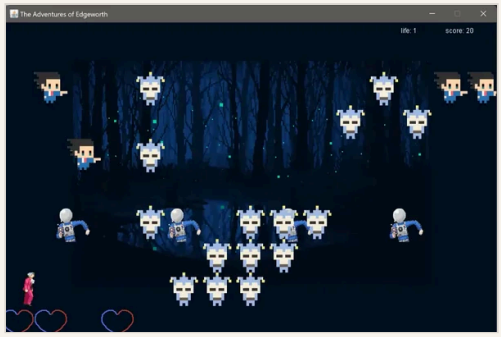
```
//Cannot move to the right
if(ob.getX()==this.x+60 &&
        (ob.getY()+50>this.y && ob.getY()-50<this.y)){
    if(ob.getType() != 3){
        canRight = false;
    }
}
```

From Edgeworth.java (one line wrapped to fit). Type 3 is the hidden block, which you can walk through until it is revealed.

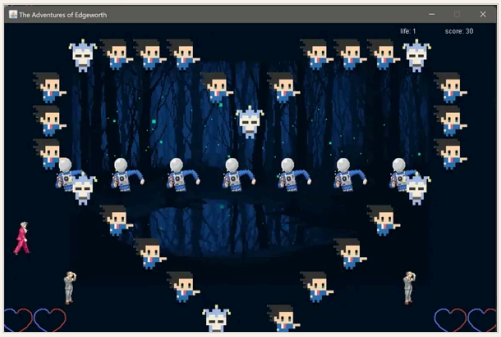


The level plans in Excel.

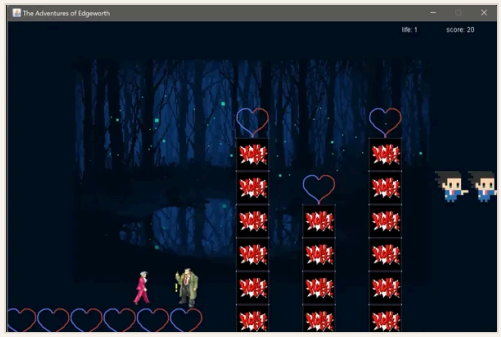
Every scene is a 15 × 10 grid of 60-pixel tiles, planned in a spreadsheet one colored cell per block. Scene 3 is a face, scene 5 a heart; scene 4 can't be finished by jumping alone, until Gumshoe's hidden "Objection!" block appears.



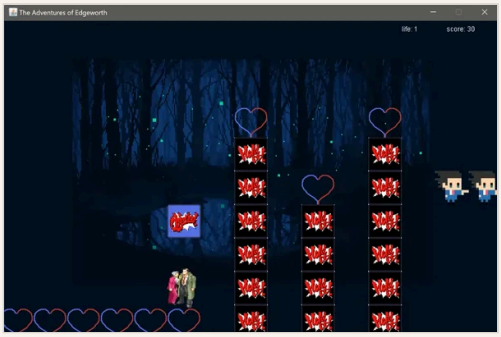
Scene 3, "the face."



Scene 5, a heart.



Before: Gumshoe waits.



After: the climb begins.

Web edition · 2026

Ported to JavaScript and the HTML canvas with AI coding tools, keeping the same coordinates, speeds and collision checks. The port found three bugs in the original, now fixed in both versions, and a breadth-first search over every reachable position proves each scene can still be finished.

05

FULL-STACK WEB · CPS 3500 · FEBRUARY 2026

AI Recipe Remix

From leftovers to plated recipes

A web app that turns whatever is in your kitchen into three recipes, each with a photo of the finished dish. React and Tailwind on the front end; a small Express server talks to OpenAI, so the API key never reaches the browser.

ROLE

Solo: front end, back end, prompts. The 2026 redesign was built with AI coding tools.

TIMELINE

February 2026

CONTEXT

CPS 3500 Project 1, Kean University

BUILT WITH

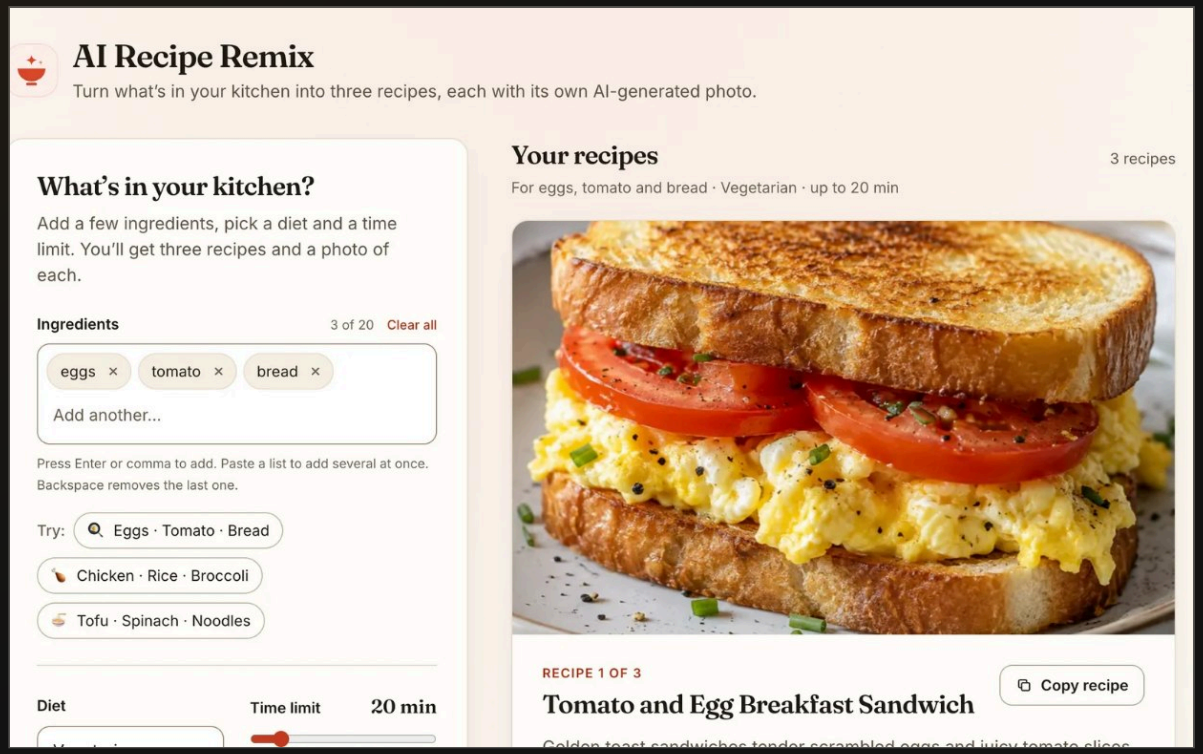
React, Vite, Tailwind CSS, Express, OpenAI API

Try the demo

samsi0322.github.io/xinyue-feng/play/recipe-remix

Case study

samsi0322.github.io/xinyue-feng/work/recipe-remix



The app after a run with eggs, tomato and bread.

From the fridge to the plate



Skillet Tomato Eggs with Toast Soldiers



Garlic Soy Chicken and Broccoli Rice Bowls



Tofu, Spinach, and Mushroom Noodle Soup

The idea

Most recipe sites start from a dish; this one starts from your fridge. Type what you have, choose a diet and a time limit, and it suggests three realistic recipes with quantities, steps, substitutions, allergens and tips.

How it works

Two requests, kept apart on purpose: recipe text comes back in seconds and the photos follow. The server asks the model for JSON in a fixed shape, and holds the OpenAI key so the browser never sees it.

The public demo

A paid key on the open web would invite abuse, so the demo replays pre-generated results for three sample kitchens, in exactly the format the server returns.

Portfolio edition · 2026

Revisited with AI coding tools: a new interface, “in your kitchen” vs. “to buy” lists, photos that load independently, a safer server and automated tests for both halves.

The dishes above are from the demo, generated in October 2026 by OpenAI models from the app’s own prompts.

06

GAME AI · UNIT 10 “PLAY WITH AI” · APRIL 2026

Tic-Tac-Toe vs. AI

Three opponents, one unbeatable

A small browser game made for a class unit on playing against AI, since extended into a side-by-side comparison of three opponents: one that plays loosely, one that follows five fixed rules, and one that searches ahead to the end of the game and never loses.

ROLE
Solo: design and code. The 2026 extension was built with AI coding tools.

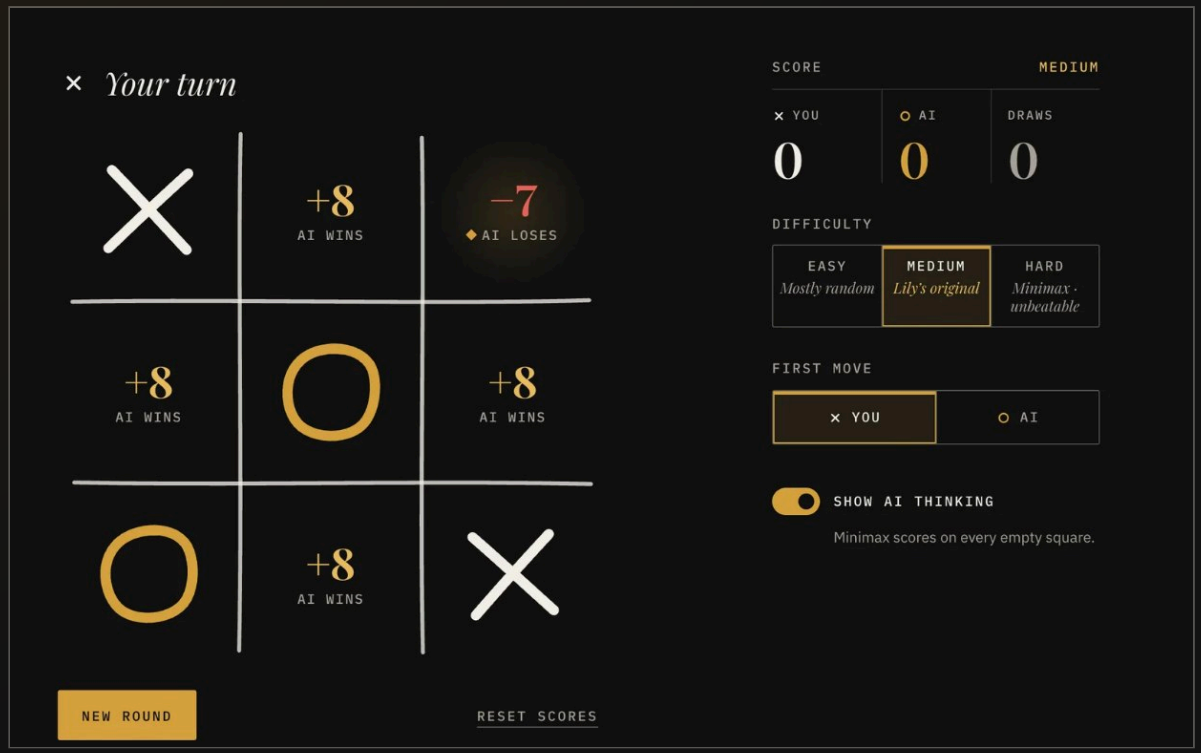
TIMELINE
April 2026, extended for this portfolio

CONTEXT
Unit 10 class activity, “Play with AI”

BUILT WITH
HTML, CSS, JavaScript; no libraries

▶ Play
samsi0322.github.io/xinyue-feng/play/tic-tac-toe

Case study
samsi0322.github.io/xinyue-feng/work/tic-tac-toe



“Show AI thinking” scores every empty square.

Rules versus search

The original

For the class activity, Lily built an opponent with five rules, checked in order:

1. Win if it can.
2. Block you if you're about to win.
3. Take the center.
4. Take a free corner.
5. Otherwise, take a side.

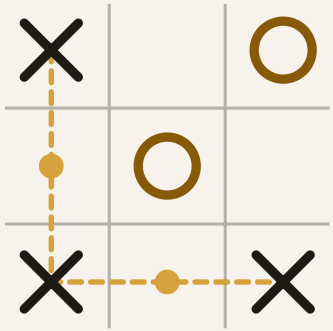
It plays sensibly and can be beaten: take a corner, then the opposite corner, and your block creates a fork it can't answer.

Three opponents

Easy mostly picks a random free square, but grabs a winning move about half the time.

Medium is her original rule-based AI, kept rule for rule; a test checks it makes the same move as the untouched original in every reachable position.

Hard uses minimax: it searches to the end of the game, assumes you always play your best, and picks the best guaranteed result. Alpha-beta pruning keeps each search to a few milliseconds.



Beating Medium: take a corner, then the opposite one. Medium takes the center, then a random corner; block the line it threatens (here, bottom left) and you have two lines that each need one more X.

Seeing the AI think

With "Show AI thinking" on, every empty square shows the minimax verdict for it: a forced win, a draw or a square that would let you win. You can watch a move turn from draw to loss the moment you set up a fork.

16,752

Possible games played against Hard in the tests. It loses none.

The 2026 extension (Easy and Hard, the overlay, a redesign, keyboard play and tests) was built with AI coding tools; her rule-based AI and the original files are kept in the repository.

CONTACT

Say hello

lily.china@outlook.com

samsi0322.github.io/xinyue-feng

PLAY

Yuzu Onsen Capybaras, winter demo
capybara.sam-si.cn

The Adventures of Edgeworth
samsi0322.github.io/xinyue-feng/play/edgeworth

Tic-Tac-Toe vs. AI
samsi0322.github.io/xinyue-feng/play/tic-tac-toe

AI Recipe Remix demo
samsi0322.github.io/xinyue-feng/play/recipe-remix

Wizarding World on GitHub
github.com/SamSi0322/WizardingWorld

This portfolio and its website were built with AI coding tools, and each project says how it was made. Fan projects are non-commercial and not affiliated with the rights holders: *Ace Attorney* © Capcom; *Harry Potter* © Warner Bros. Entertainment / J.K. Rowling; *Terraria* © Re-Logic; *Hollow Knight* and *Silksong* © Team Cherry; *Dead Cells* © Motion Twin; *Cuphead* © StudioMDHR; *The Binding of Isaac* © Edmund McMillen.